

HOW TO USE

KARBHARI

Working Capital Guardian · project guide

Run it, give it a case, and read what it finds. Every screenshot in this guide was captured from a live run of the real app on the test files that ship with the project.

1

Start the app

2

Open a case and attach
evidence

3

Run the investigation

4

Read, verify, act

STEP 1 • START THE APP

Two ways to run it.

One API key.

KARBHARI is one FastAPI backend serving a React console. Run it as a single Docker container, or run backend and frontend separately while developing. Either way you need a Google AI Studio (Gemini) API key; the free tier is enough.

Before you start: the API key

Copy `backend/.env.example` to `backend/.env` and fill it in. Keep this file out of git.

```
GOOGLE_API_KEY=your-key-from-aistudio.google.com/apikey
LLM_PROVIDER=gemini
LLM_MODEL=gemini-flash-lite-latest # an alias, tracks Google's current model
```

OPTION A • RECOMMENDED

Docker: the whole product in one container

```
# build from the karbhari/ folder
docker build -t karbhari:0.7.0 .
run.bat

# or use the published image
docker run -p 8000:8000 \
  --env-file backend/.env \
  yeldnever/karbhari:0.7.0
```

`run.bat` replaces any old `karbhari` container and starts the new one. Open <http://127.0.0.1:8000>.

OPTION B • DEVELOPMENT

Backend + frontend, two terminals

```
# 1 • backend (Python 3.12+)
cd backend
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
uvicorn app.main:app --port 8000

# 2 • frontend (Node 20+)
cd frontend
npm install
npm run dev
```

Open <http://127.0.0.1:5173>. Vite forwards `/api` to port 8000.

Check it is alive

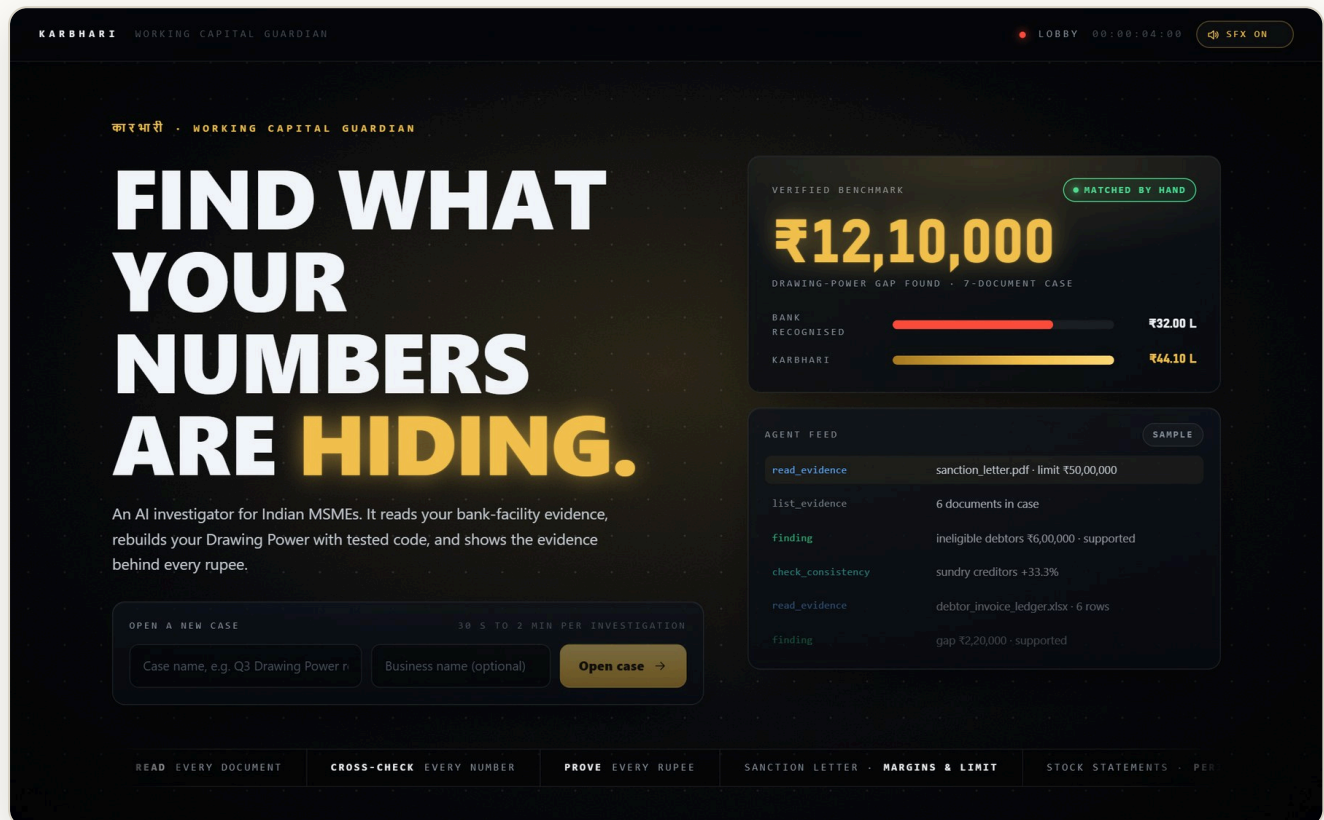
```
curl http://127.0.0.1:8000/api/health
→ {"status":"ok","service":"karbhari-backend"}
```

Data lives in `backend/data/` (SQLite database and uploaded files). Set `KARBHARI_DATA_DIR` to keep a separate workspace, for example a clean one for demos.

STEP 2 • OPEN A CASE

One case per review.

The lobby is where every investigation starts. Type a case name (for example *Q3 Drawing Power review*), add the business name if you like, and click **Open case**. Your cases appear further down the page; click one to open its workspace.



THE LOBBY • OPEN A CASE, SEE THE VERIFIED BENCHMARK AND A SAMPLE OF THE AGENT AT WORK

ON THIS SCREEN

- **Verified benchmark:** the ₹12,10,000 gap KARBHARI matched to a hand-worked answer.
- **Agent feed:** an illustrative sample of the steps the agent takes (labelled *sample*).
- **How it works** and **what it catches**, further down.

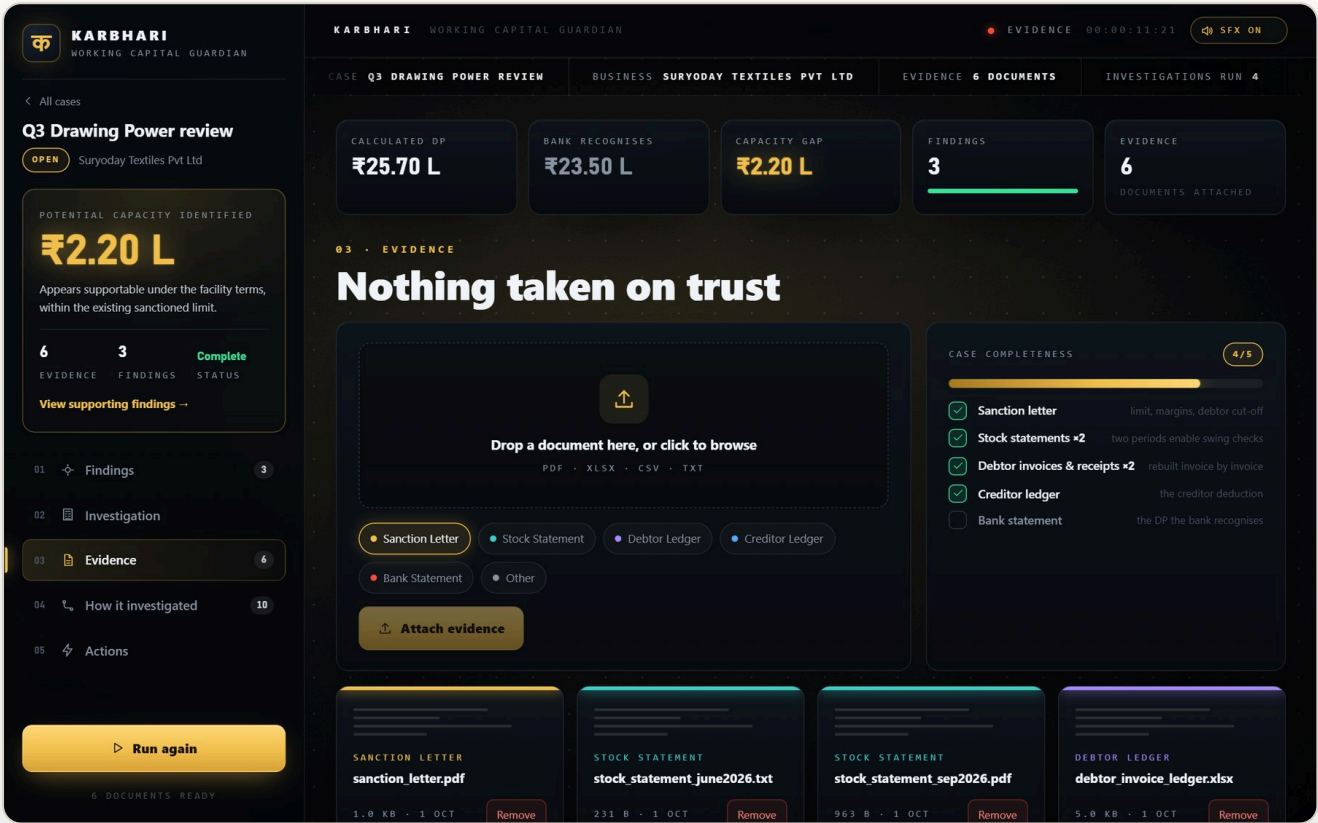
SOUND AND MOTION

The console uses subtle interface sounds: clicks, a whoosh between tabs, a chime when something is saved and an impact when a result lands. Toggle them with **SFX ON** in the top-right corner; your choice is remembered.

STEP 2 • ATTACH EVIDENCE

Tag every file.

In the case, open **Evidence**. Drop a file on the upload area (or click to browse), pick its category chip, then click **Attach evidence**. Repeat for each document. PDF (with a text layer), XLSX, CSV and TXT all work.



EVIDENCE • THE SIX TEST FILES FROM ASSETS FOR TESTING, EACH TAGGED, WITH THE COMPLETENESS CHECKLIST

CATEGORY	UPLOAD	GIVES KARBHARI
sanction letter	The bank's sanction / facility letter	Limit, margins, debtor age cut-off
stock statement	Monthly stock statements (attach two months if you have them)	Stock value; period-over-period check
debtor ledger	Invoice-level ledger and receipts / payment records	Eligible debtors, rebuilt invoice by invoice
creditor ledger	Sundry creditors list	The creditor deduction
bank statement	CC account statement showing the DP the bank recognises	The figure to compare against

The **Case completeness** checklist ticks off each category as you attach it, so you can see at a glance what is still missing before you run.

STEP 3 • RUN THE INVESTIGATION

Click once. Watch it work.

Click **Run investigation** at the bottom of the left rail. A full-screen view shows the agent reading each of your documents, cross-checking them and calculating Drawing Power. A run usually takes **30 seconds to 2 minutes**, depending on the model's response time.



INVESTIGATING • A GOLD BEAM TICKS OFF EACH REAL DOCUMENT WHILE THE LIVE LOG SHOWS THE CURRENT STEP

WHAT HAPPENS BEHIND THE SCENES

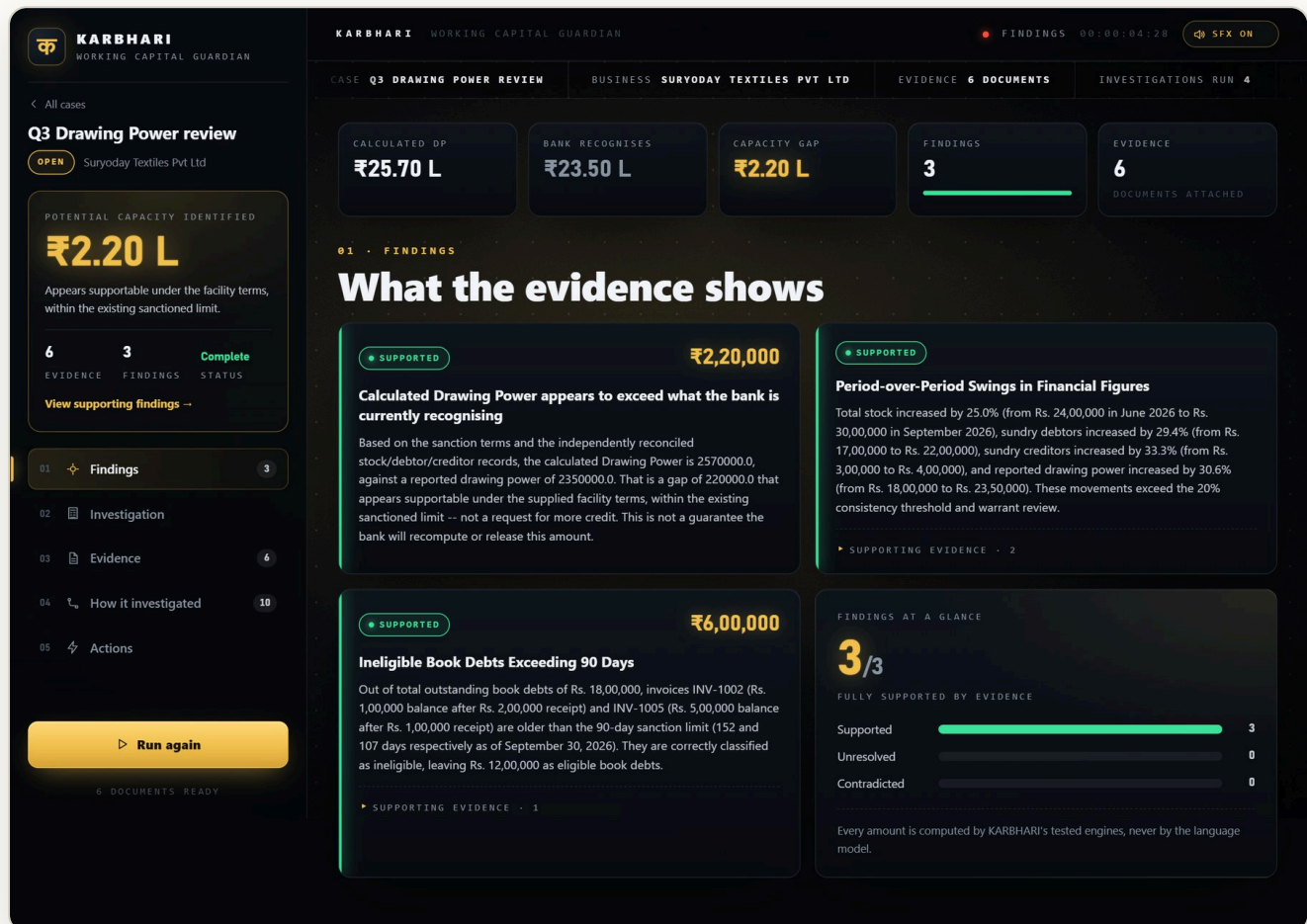
`list_evidence` → `read_evidence` on every file →
`check_consistency` → `reconcile_debtors` →
`calculate_drawing_power` → findings. Bounded at 20 steps. Every number comes from tested Python, never from the language model.

WHEN IT FINISHES

The overlay lifts, the headline figure in the left rail counts up with gold rings, and you land on **Findings**. If the model is slow, the overlay says so after 30 seconds; nothing is lost.

STEP 4 • READ THE FINDINGS

The headline, then the reasons.



FINDINGS • LIVE RESULT ON THE BUNDLED TEST CASE

The left rail and top strip

- **Potential capacity identified:** the Drawing Power gap, here **₹2.20 L** (calculated ₹25,70,000 vs ₹23,50,000 recognised by the bank).
- The **KPI strip** repeats the key figures on every tab.
- **Run again** re-runs after you add or change files.

How sure each finding is

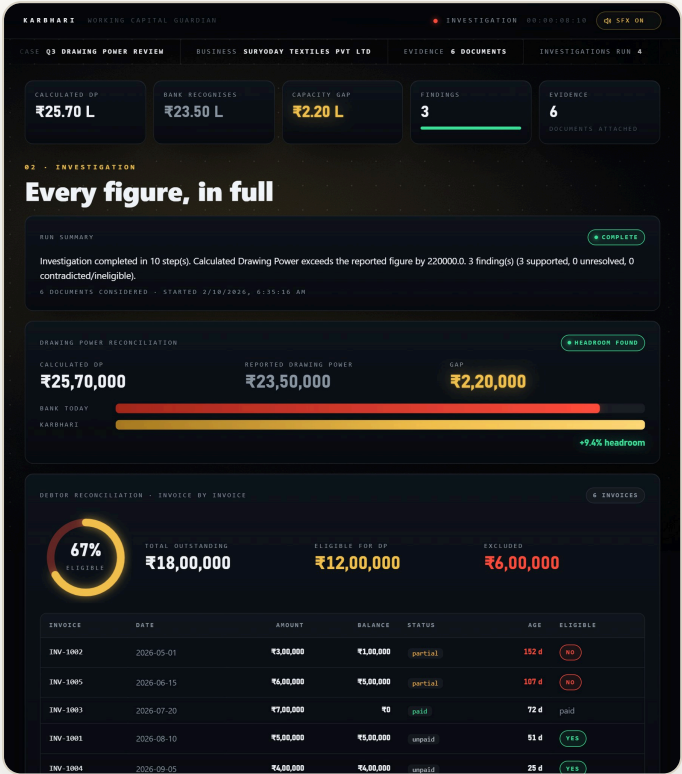
- **SUPPORTED** evidence and a tool output directly back it.
- **UNRESOLVED** incomplete or ambiguous, or an assumption was needed.
- **CONTRADICTED** the evidence works against the claim.
- Open **Supporting evidence** on any card for the exact quotes.

The gap is headroom **within the existing sanctioned limit** that appears supportable under the facility terms. It is not a request for more credit, and not a guarantee the bank will release it.

STEP 5 • CHECK THE NUMBERS

Every figure, shown in full.

Open **Investigation** to see the working behind the headline.



INVESTIGATION • RECONCILIATION DETAIL FROM THE LIVE RUN

DRAWING POWER RECONCILIATION

Calculated DP, the bank's figure and the gap, drawn as two capacity bars with the headroom percentage. Any assumption the engine had to make is listed; if inputs are missing, it names them.

DEBTOR RECONCILIATION

The eligibility ring shows how much of the debtor book counts towards Drawing Power. Below it, every invoice with balance, status, age and eligibility. In the test case, INV-1002 (152 days) and INV-1005 (107 days) are past the 90-day cut-off, so **₹6,00,000** of ₹18,00,000 is excluded.

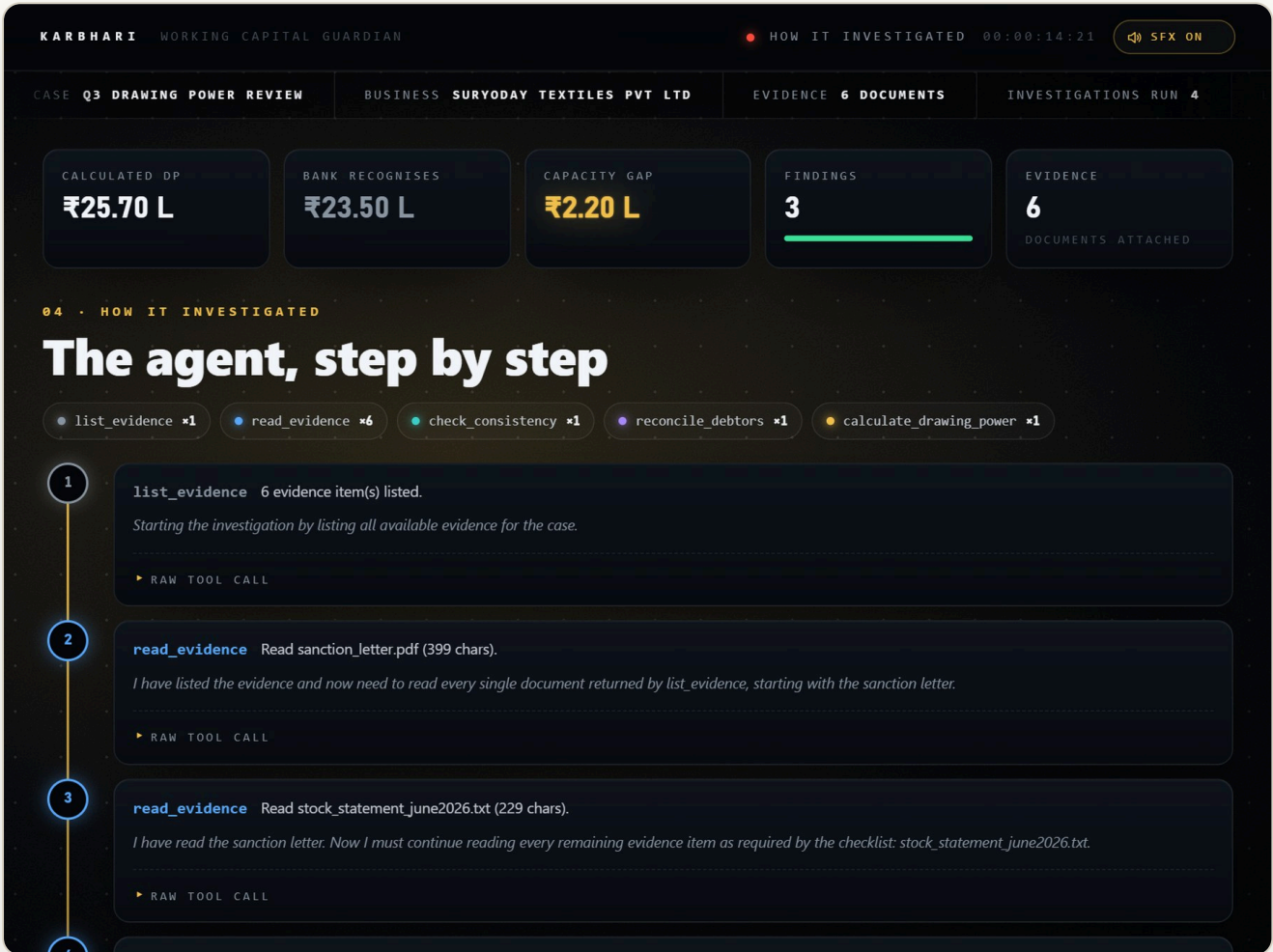
PERIOD-OVER-PERIOD VARIANCE

Any figure that moved 20% or more between statements: here stock +25.0%, debtors +29.4% and creditors +33.3% from June to September. Confirm what drove them before relying on the latest month.

STEP 6 • AUDIT THE REASONING

See exactly how it got there.

Open **How it investigated** for the step-by-step timeline of the live run: the agent's reasoning, the tool it called and a summary of the result. Expand **Raw tool call** to see the exact input and output.



HOW IT INVESTIGATED • THE REAL TRACE, RECORDED AS IT RAN

READING THE TIMELINE

Each tool has its own colour, and the pills at the top count how often it ran: one `list_evidence`, one `read_evidence` per file, then the checks and the calculation.

ACTIONS TAB

Previews what comes next: a draft note to your bank, an evidence pack PDF, a follow-up checklist and a monthly watch. These are marked *next phase*; for now, take the findings and the Investigation tab to your banker or CA.

TIPS AND TROUBLESHOOTING

Getting a complete answer.

IF YOU SEE...

DO THIS

UNRESOLVED Drawing Power was not calculated	The agent finished before running the calculation. It is non-deterministic, so click Run again ; each run is kept in the case history.
...could not be independently calculated. Missing: ...	Upload the document named in the message: usually the sanction letter, a current stock statement or a debtor ledger.
...nothing to compare it against	Add a bank / CC statement (or a document stating the DP the branch recognises).
Placeholder margins / creditors treated as zero	The sanction letter or creditor ledger wasn't found or readable. The result is an estimate until you add them.
Consistency was not checked in this run	Two same-category statements exist but the agent skipped the comparison. Re-run, or compare the periods yourself.
"No extractable text found"	The PDF is a scan. There is no OCR yet; upload a text PDF, Excel or CSV export instead.
"Taking longer than usual"	The model is slow right now. Leave the overlay open; the result appears when the run completes.
Investigation could not start	Check <code>GOOGLE_API_KEY</code> in <code>backend/.env</code> and restart.
Port 8000 already in use	Another KARBHARI (often the Docker container) is running. Stop it, or start <code>uvicorn</code> with <code>--port 8010</code> .
No sound	Check <code>SFX ON</code> in the top-right. Browsers only play audio after your first click on the page.

FOR THE BEST RESULT, UPLOAD

- The sanction letter, with margins and the debtor age limit
- Two consecutive stock statements
- Invoice-level debtors *and* receipts, not just a subtotal
- A creditor list and the bank's recognised DP

TRY IT NOW

The six files in `karbhari/ASSETS FOR TESTING/` reproduce the run in this guide: a **₹2.20 L** gap and 3 supported findings.

Run the tests: `cd backend` then `python -m pytest -q`.